LEXTRIA RESEARCH

An Astraea Innovative Solutions, #21, 4th floor, MUTBI, ARC, Manipal - 576104

14-10-2026

DUE DILIGENCE REPORT: AR1114

Dear Inventor,

We have completed our due diligence under the Computer Related Inventions (CRI) Guidelines issued by the Hon. Indian Patent Office, as instructed by Hon. IPTTO MAHE, and we set out our observations below in respect of the invention titled "Deterministic Multi-Signal Processing for Academic Integrity Verification with Missing-Evidence Bounds and Cryptographically Linked Phase Records", being a computer-implemented system that evaluates venue, citation, peer-review and publication-policy records and outputs a risk status with missing-evidence bounds and hash-linked phase records. We set out our observations in the hope that they assist you, and we would be grateful for your help on the points raised below.

1. SECTION 3(k) OBJECTION RISK: MATHEMATICAL METHOD, ALGORITHM AND COMPUTER PROGRAMME PER SE

We are of the view that the disclosure, as it presently stands, faces a credible risk of objection under Section 3(k) of the Patents Act, 1970. In our assessment, an examiner is likely to find that the substance of the disclosure lies in mathematical computation and rule-based logic executed on a general-purpose computer, with the output being a score, an interval and a risk tier.

The disclosure's researcher auditing module enumerates directed citation cycles, partitions citations by precedence and recomputes the h-index, CIR and LACIR. Each result is an index or a ratio. The review module computes sentence-length variation, hedging density, specificity and polarity, and the venue module computes edit distance, a deadline interval and a weighted root-mean-square fingerprint change. An examiner is likely to treat each of these as a mathematical method whose end goal is the number itself.

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The disclosure's observed-set weighting with bounds and its cross-signal amplification are claimed as formulas. We expect an examiner to read these claimed formulas as a mathematical method and an algorithm presented in the form of a system.

The claimed system refers to processors and memory only generically, the claimed method of processing records makes no reference to any hardware, the claimed storage medium holds instructions for performing that method, and the claimed display apparatus and submission gateway recite only a processor, memory, a display or a network interface as hardware. Under the CRI Guidelines the examiner looks to the substance of the invention and not to the form of the claim, and we are of the view that these claims are likely to be treated as a computer programme per se.

The Patent Specification Report presents the exclusion of machine learning and a "mathematically verifiable" pipeline as the distinguishing attribute, in its abstract, its discussion of the separation between the offline sandbox and the production pipeline, and its description of the end-to-end method, where the result is described as a "100% mathematically verifiable evidence chain". In our assessment, this describes how the system is designed and is not by itself a technical effect. The emphasis on explicit formulas and Boolean logic may also draw the examiner's attention towards the exclusion.

The performance statements are the processing of 11,500 synthetic inputs (1,000 citation graphs, 10,000 publication portfolios and 500 text samples) within a stated time and a 100% cartel identification result on synthetic graphs, as given in the Patent Specification Report and reported again in the Specification. The disclosure does not state the hardware, a baseline for comparison or an effect on a technical metric, and the Specification itself notes that the absence of execution errors is distinct from correct classification. An examiner is likely to ask what technical improvement results, and measured against what.

The thresholds appear as bare values: 0.35, 0.60, 0.20, 0.50, 0.30, 0.25, 0.04, 0.15, 30 days and a cycle length of five. Several are described in the Specification itself as illustrative, as requiring confirmation by the inventors, or as yet to be supplied, including the indexing score mapping, the

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hedging lexicon, the specificity denominator, the polarity formula and the non-facade amplification factor. An examiner may regard such values as arbitrary unless the reason for each value is stated.

We are of the view that this objection is best answered where the disclosure shows a technical effect beyond the programme itself, for an identified technical problem and by identified technical means.

Input required from the inventor: We would be grateful for measured results with and without each of the following features: the observed-set weighting with bounds, the cross-signal amplification and the hash-linked phase records. For each, please state the metric used, the test set, the hardware and the baseline against which the result is compared. We would also be grateful if, for each threshold named above, you could let us know the value, the reason for that value, and the technical consequence of a value below or above it.

2. SECTION 3(k) OBJECTION RISK: BUSINESS METHOD

We are of the view that the disclosure, as it presently stands, also faces a credible risk of objection under the business method limb of Section 3(k). The Section 3(k) exclusion for a business method is not qualified by "per se", and an examiner is likely to ask where the core of the invention lies.

The disclosure's grant compliance module selects a publication policy by authority and effective date, tests publication metadata such as open-access route, repository deposit, embargo duration, licence and funding acknowledgment against that policy, and maps each failed outcome to a remediation code. In our assessment, an examiner is likely to characterise this as the application of administrative rules to a record, which is a compliance procedure carried out by computer. The disclosure itself states that the sample policies are synthetic and that a failed predicate is a result under a chosen rule version and not a legal determination, and an examiner may take that as confirming that the module's contribution lies in the rules and not in the machinery.

The submission gateway receives a submission, assesses it and returns a release, hold or rejection code according to a risk tier and an evidence-coverage rule, as claimed. The claimed display

apparatus presents the same assessment on a display. We expect an examiner to treat the release, hold or rejection decision as an editorial or institutional decision on a manuscript, and the display as the presentation of information to a user. The disclosure does not identify any technical system component whose operation is controlled or improved by the action code.

The same reasoning is likely to be applied to the assessment of venue credibility, researcher independence and peer-review authenticity, since the disclosure presents the system as an academic integrity assessment used through an institutional dashboard and a manuscript submission gateway.

We are of the view that this objection is answered where the business context is shown to be only a constraint on scope, and the contribution is shown to be a technical improvement to an underlying system or process. In our assessment, the features of the disclosure that may be argued in this direction are the evaluation of policy predicates with an explicit unknown state, the selection of a policy version by effective date so that a past assessment can be reproduced, and the coverage rule that prevents an adverse automated action on incomplete evidence. These can be presented as data-handling measures only if the disclosure shows what technical problem each solves and what measurable effect results.

Input required from the inventor: We would be grateful if you could let us know what is technically improved, if the funder, publisher and institutional context is set aside, by the policy-version selection, the three-valued predicate evaluation and the coverage-based action rule, with a measured result where available. We would also be grateful if you could describe the system component that receives the release, hold or rejection code and the operation that component performs on receipt, for example the routing or locking of a submission record in a submission platform.

3. POINTS WHERE THE DOCUMENTS DIFFER

In reviewing the Patent Specification Report, the Specification and the Technical Figure Set together, we noted the points below where the documents differ in a value or description, or where

one document describes a feature that another does not. We would be grateful for your confirmation of what is intended in each case, so that the filing reflects the invention as implemented.

- The Patent Specification Report removes self-citations and co-author citations to obtain independent citations, the True Independent H-Index and the Citation Independence Ratio, with an alert below 0.30. The Specification describes a further loop-associated category, a loop-purged h-index and a loop-adjusted ratio, each with an alert below 0.30. It uses "independent citations" for the category remaining after loop removal, so the Citation Independence Ratio in the Specification also counts loop-associated citations.

- The Patent Specification Report flags a review where polarity exceeds 0.6 with a Reject recommendation. The Specification flags polarity of at least 0.50 with Reject and of at most −0.30 with Accept.

- The Patent Specification Report gives the formula for hedging density, expressed as a percentage of total words, and describes specificity by regular-expression matching, without a threshold for either. The Specification gives a hedging threshold of 0.04 as a fraction of word tokens and a specificity threshold of 0.15, and defines the denominators of both.

- The Patent Specification Report states that the bounded search for citation rings runs in linear time in the number of authors and citations. The Specification states that no linear bound is asserted for complete cycle enumeration, and describes a canonical minimum-node start with a recorded truncation status.

- The Patent Specification Report states 100% cartel identification with zero false negatives and processing of 11,500 inputs in under one second. The Specification states a processing time of 0.589 seconds, seven verification suites and nine completed examples. The ground truth and the cycle lengths covered by the 100% result, and the run to which the 0.589 second figure relates, were not included in the material we received.

- The Patent Specification Report groups Adaptive Decay Weighting and Cross-Signal Anomaly Detection in a single credibility phase. The Specification places the weighting in component scoring and the anomaly detection in anomaly aggregation. The weighting redistributes weights proportionally and contains no time-decay function.
- The Patent Specification Report presents machine-readable rule matrices for named funders with an example rule. The Specification states that no current policy requirement is hardcoded and uses versioned policy records selected by effective date.
- The Patent Specification Report describes an on-chain deterministic pipeline and an off-chain sandbox. The Specification describes hash-linked phase records verified against a retained anchor and states that no blockchain is required.

Input required from the inventor: We would be grateful for your confirmation, for each point above, of the value or description that corresponds to the implemented system. Please also let us know whether the loop-associated category and loop-adjusted ratio are implemented, whether any time-decay function is used in the weighting, and whether any distributed ledger is used for the phase records.

4. CONSOLIDATED SUPPORT WE WOULD BE GRATEFUL TO RECEIVE

Measured technical effect (supporting Part 1)

We would be grateful for measured results with and without the observed-set weighting with bounds, the cross-signal amplification and the hash-linked phase records. For each, please state the metric, the test set, the hardware and the baseline compared.

Reasons for the thresholds (supporting Part 1)

We would be grateful for the value, the reason for that value and the technical consequence of a value below or above it, for each threshold named in Part 1.

Hardware and deployment (supporting Part 1)

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The disclosure states that the system may run as a command-line process, a service or an institutional application. We would be grateful for a description of the processor, memory, data volume and number of concurrent assessments used in the reported runs and in deployment, so that each measured result can be tied to an identified technical setting.

Policy handling and the action code (supporting Part 2)

We would be grateful for what is technically improved by the policy-version selection, the three-valued predicate evaluation and the coverage-based action rule, with a measured result where available. We would also be grateful for a description of the system component that receives the release, hold or rejection code and the operation it performs on receipt.

Implemented embodiments (supporting Parts 1 and 3)

We would be grateful for a list of the features that are implemented in the system as built, together with the complete hedging lexicon, the indexing-language patterns, the authorised host registry and the policy records used, since the disclosure gives examples only.

Confirmation of the points that differ (supporting Part 3)

We would be grateful for your confirmation, for each point in Part 3, of the value or description that corresponds to the implemented system.

Conclusion: In our assessment, the objection risks under Section 3(k), namely mathematical method, algorithm, computer programme per se and business method, remain open on the disclosure as it presently stands. Our further observations will follow on receipt of your response to the points above.

Prepared by: Lextria Research Team